

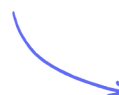
Exam Questions

Probabilities from Tables & Charts

Two-Way Tables / Possibility (Sample Space) Diagrams / Probabilities from Venn Diagrams

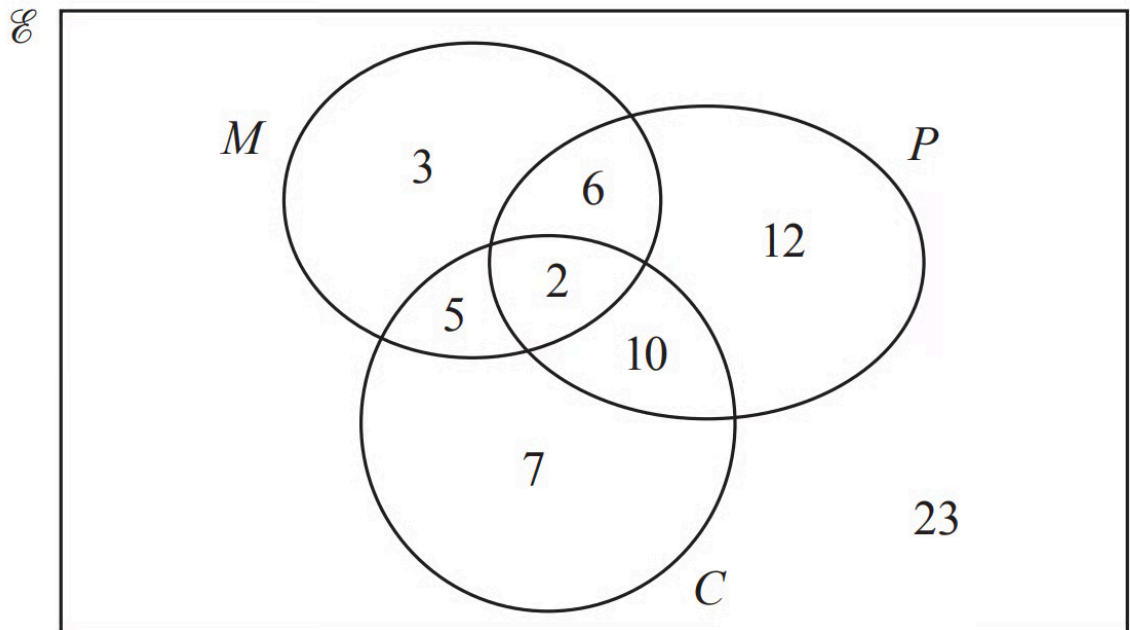
Easy (3 questions)	/5
Medium (6 questions)	/26
Hard (3 questions)	/12
Total Marks	/43

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Easy Questions

- 1 The Venn diagram below shows information about the number of gardeners who grow melons (M), potatoes (P) and carrots (C).

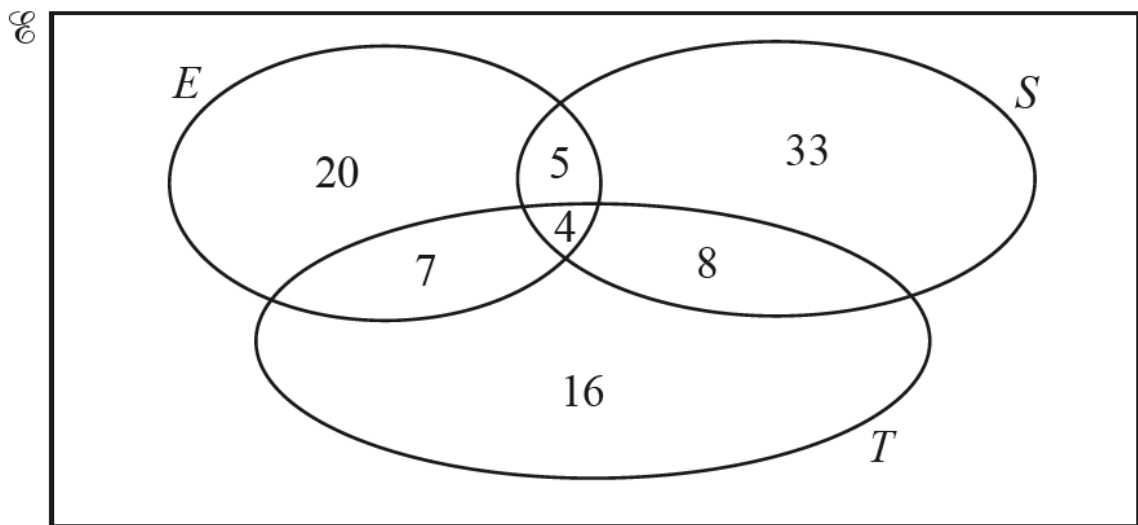


A gardener is chosen at random from the gardeners who grow melons.
Find the probability that this gardener does not grow carrots.

(2 marks)

- 2 At a fitness centre, the number of members using the exercise machines (E), the swimming

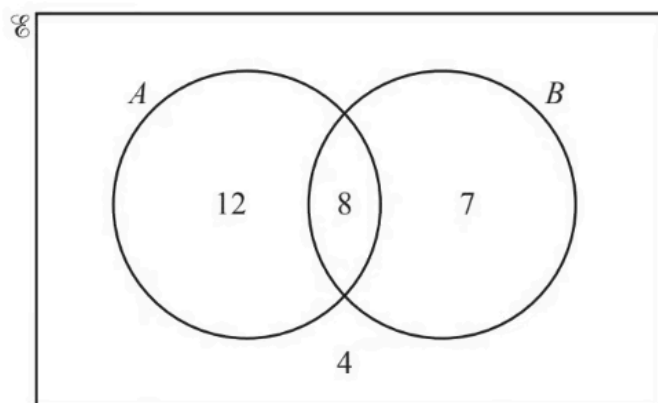
pool (S) and the tennis courts (T) is shown on the Venn diagram.



A member using the swimming pool is chosen at random. Find the probability that this member also uses the tennis courts and the exercise machines.

(2 marks)

- 3** The Venn diagram shows information about the number of students in a class who like apples (A) and bananas (B).



One of the students is chosen at random.

Find the probability that this student does not like apples and does not like bananas.

[1]

(1 mark)

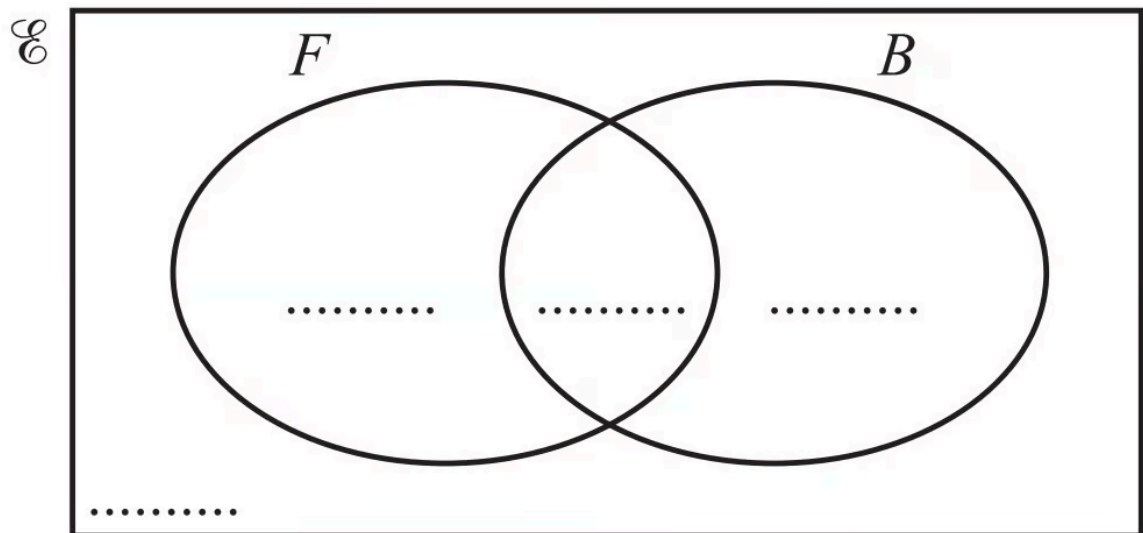
Medium Questions

- 1 (a) $\mathcal{E} = \{\text{students in a school}\}$
 $F = \{\text{students who play football}\}$
 $B = \{\text{students who play baseball}\}$

There are 240 students in the school.

- 120 students play football
- 40 students play baseball
- 90 students play football but not baseball.

Complete the Venn diagram to show this information.



(2 marks)

- (b) A student in the school is chosen at random.
Find the probability that this student plays baseball but not football.

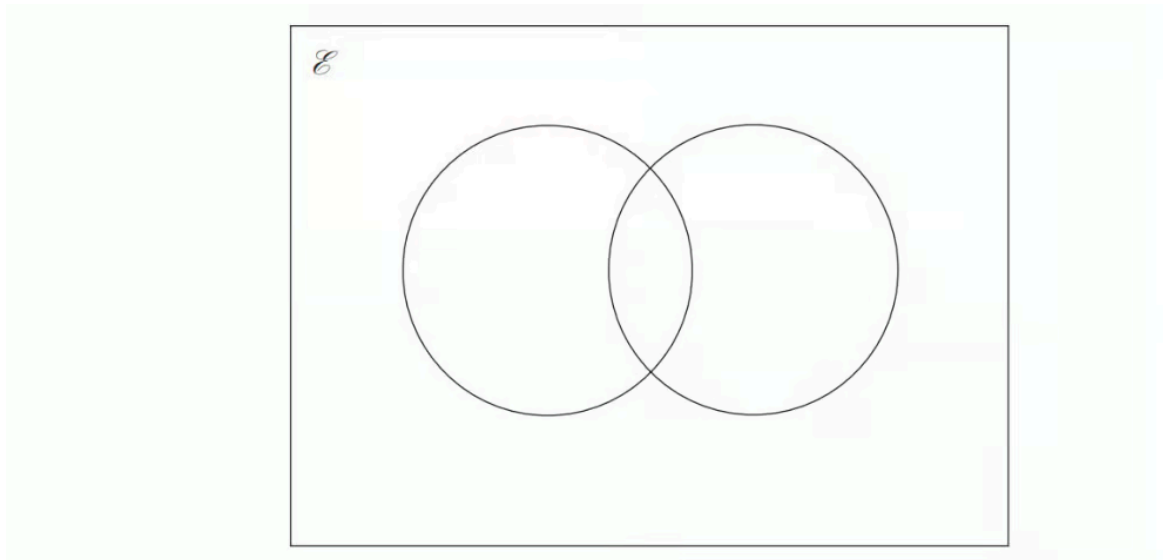
(1 mark)

2 (a) $\mathcal{E} = \{\text{odd numbers less than 30}\}$

$$A = \{3, 9, 15, 21, 27\}$$

$$B = \{5, 15, 25\}$$

Complete the Venn diagram to represent this information.



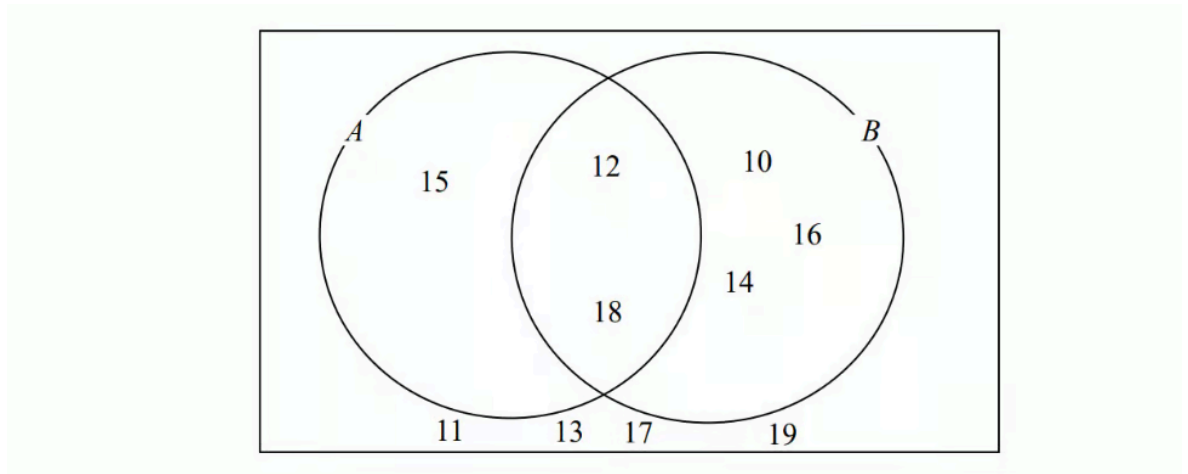
(4 marks)

(b) A number is chosen at random from the universal set, $\mathcal{E}$.

What is the probability that the number is in the set $A \cup B$?

(2 marks)

3 (a) Here is a Venn diagram.



Write down the numbers that are in set

i) $A \cup B$

[1]

ii) $A \cap B$

[1]

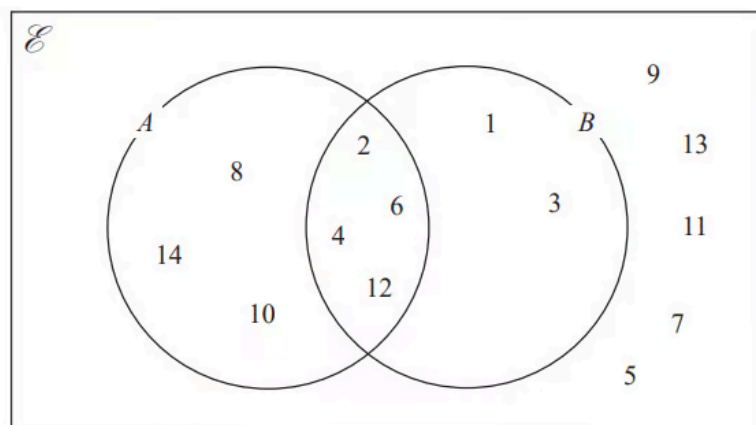
(2 marks)

(b) One of the numbers in the diagram is chosen at random.

Find the probability that the number is in set A' .

(2 marks)

4 (a) The numbers from 1 to 14 are shown in the Venn diagram.



i) List the members of the set $A \cap B$

[1]

ii) List the members of the set B'

[1]

(2 marks)

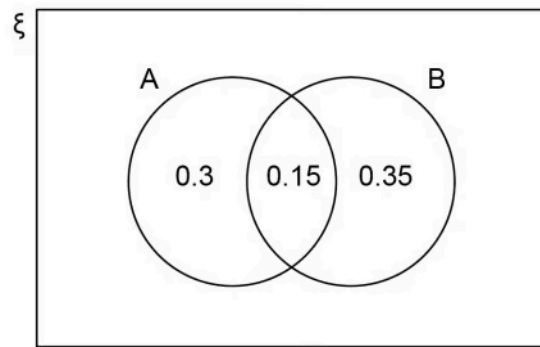
(b) A number is picked at random from the numbers in the Venn diagram.

Find the probability that this number is in set A but is **not** in set B .

(2 marks)

5 A and B are two events.

Some probabilities are shown on the Venn diagram.



Work out $P(A' \cup B)$.

(2 marks)

6 (a) In a group of 200 students:

- 120 study Chemistry (C),
- 90 study Geography (G),
- 50 study both Chemistry and Geography.

Draw a Venn diagram to represent this information.

(4 marks)

(b) A student is selected at random.

Find:

(i) $P(C \cup G)$.

(ii) $P((C \cup G)').$

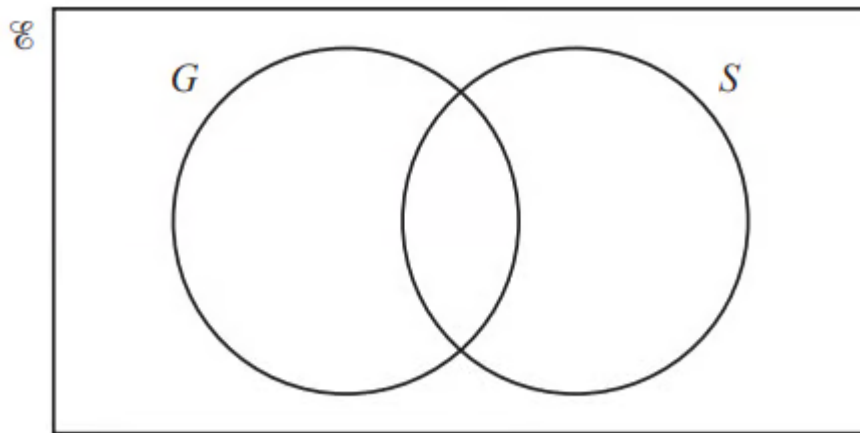
(iii) $P((C \cap G)').$

(3 marks)

Hard Questions

1 (a) There are 32 students in a class.

5 do not study any languages. 15 study German (G). 18 study Spanish (S).



Complete the Venn diagram to show this information.

(2 marks)

(b) A student is chosen at random.

Find the probability that the student studies Spanish but not German.

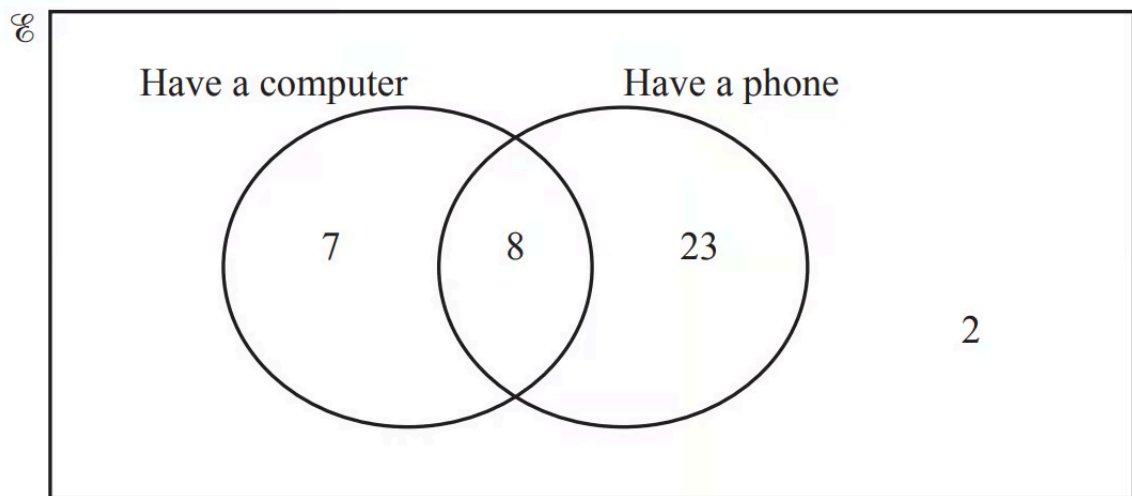
(1 mark)

(c) A student who studies German is chosen at random.

Find the probability that this student also studies Spanish.

(1 mark)

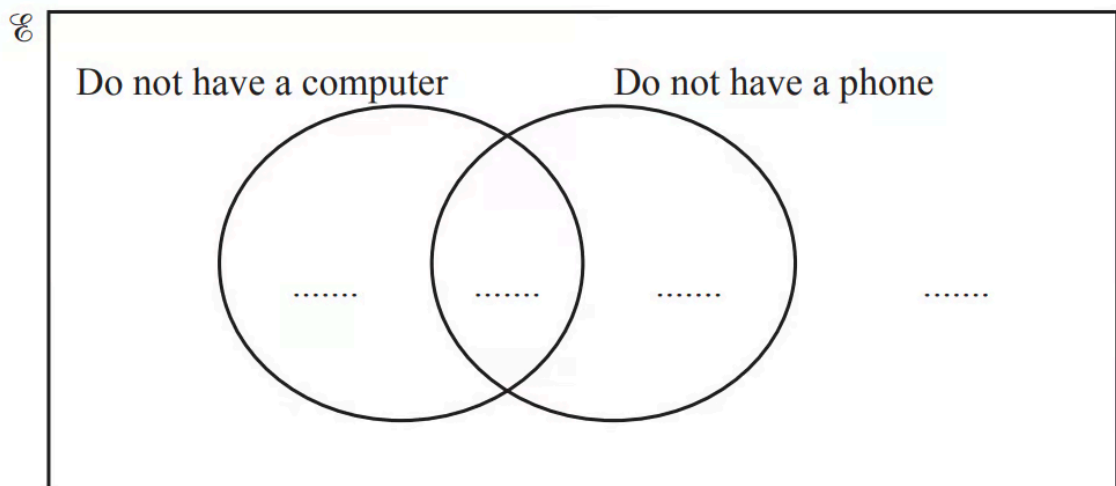
- 2 (a)** 40 children were asked if they have a computer or a phone or both.
The Venn diagram shows the results.



A child is chosen at random from the children who have a computer.
Write down the probability that this child also has a phone.

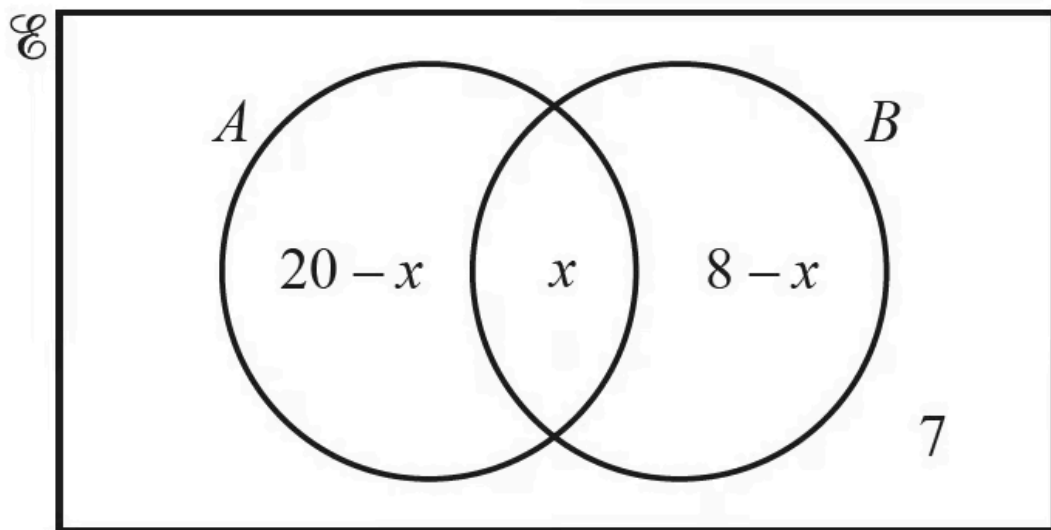
(1 mark)

- (b)** Complete the Venn diagram.



(2 marks)

3 (a) The Venn diagram shows information about the number of elements in sets A , B and $\mathcal{E}$.



$n(A \cup B) = 23$ Find the value of x .

$x = \dots\dots\dots$

(3 marks)

(b) An element is chosen at random from $\mathcal{E}$.
Find the probability that this element is in $(A \cup B)'$.

(2 marks)